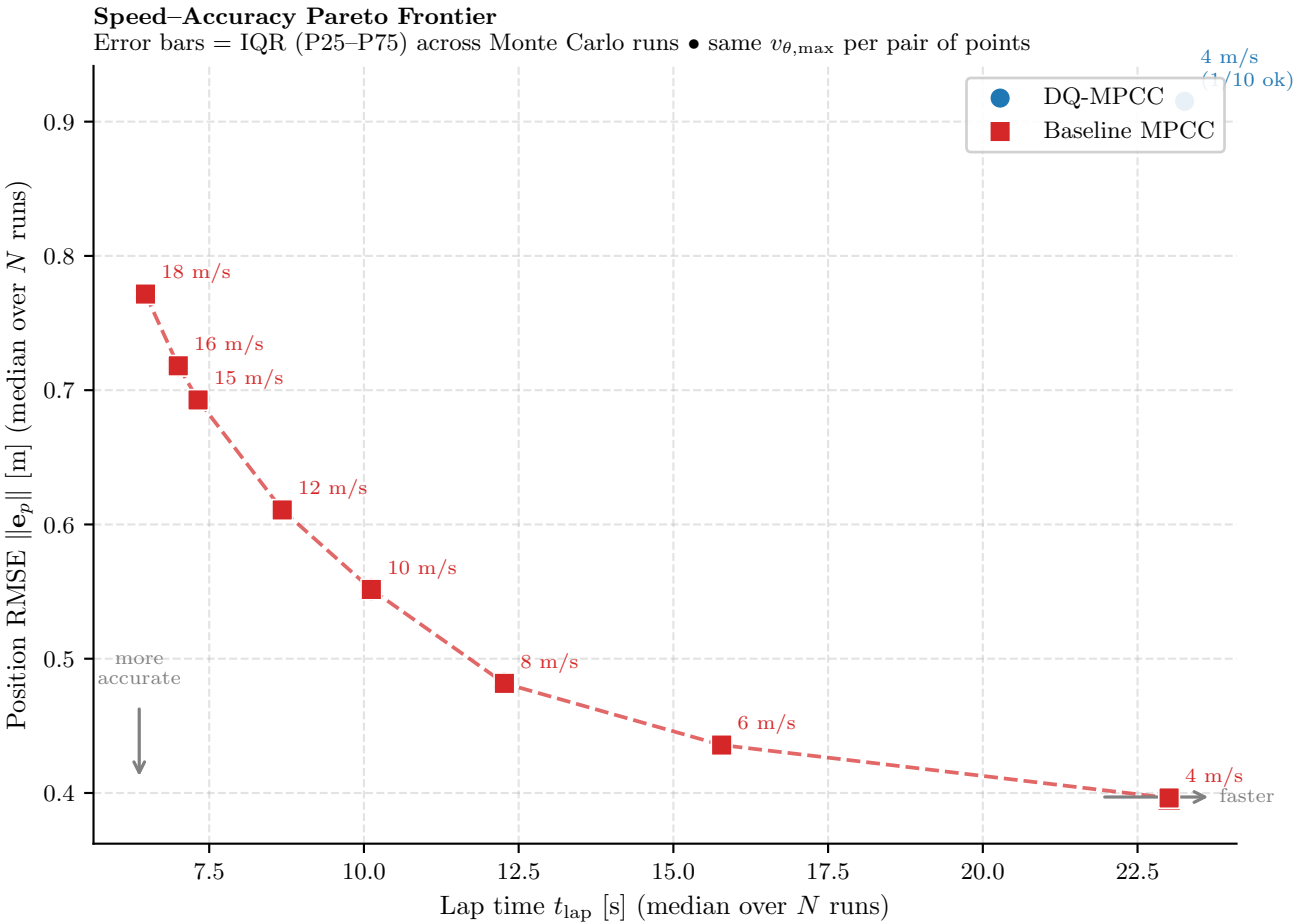
Speed–Accuracy Pareto Frontier
Error bars = IQR (P25–P75) across Monte Carlo runs • same $v_{\theta,\max}$ per pair of points
DQ-MPCC
Baseline MPCC
4 m/s
(1/10 ok)
18 m/s
16 m/s
15 m/s
12 m/s
10 m/s
8 m/s
6 m/s
4 m/s
more accurate
faster
Position RMSE $\|\mathbf{e}_p\|$ [m] (median over N runs)
Lap time t_{lap} [s] (median over N runs)
0.9
0.8
0.7
0.6
0.5
0.4
7.5
10.0
12.5
15.0
17.5
20.0
22.5